

Title: Artificial intelligence in healthcare and medicine: clinical applications, therapeutic advances, and future perspectives.

Summary:

This picture shows that AI is doing everything in healthcare systems. It has the ability to predict analysis, drug discovery and so on. It can also analysis complex data types like electronic health records, medical image, identify patterns, disease diagnostic and so on. This description also talks about frameworks to ensure patient safety, accountability and transparency. AI is also a powerful tool for innovation, sustainability and equity for healthcare system. But most of the case healthcare system can face some issues like data privacy, limited access to care and so on. There is also a solution to solve these problems. There will provide a tool where it can convert healthcare delivery through complex data and decision support. ML can reach out large amount of data such as medical history, patient data, result, clinical support etc. ML has 3 types of data Supervised, Unsupervised and Reinforcement data. Deep learning algorithm can detect abnormalities like X-rays, CT-scan etc. AI also can detect breast cancer, lung cancer and so on. Another picture discuss robotic surgery, pharmaceutical manufacturing and AI-based nanorobots. It shows that AI is not limited to diagnosis. It is moving into many parts of modern healthcare. At the same time there is talking about AI that should not replace instead it should support human judgement. AI will not only optimize clinical work flows but also reshape healthcare delivery systems on a global scale. At last the future of AI in healthcare will depend on diverse training data, stronger regulation, clinical education, cost-effective tools. That's why AI can improve healthcare outcomes and more accessible.

Link: <https://doi.org/10.1186/s40001-025-03196-w>